That is,

$$A=[a_{ij}]_{m\times n}=\begin{array}{c} \xrightarrow{\quad n \text{ columns} \quad} \\ \begin{matrix} \text{Column 1} & \text{Column 2} & & \text{Column } j & & \text{Column } n \\ \downarrow & \downarrow & & \downarrow & & \downarrow \end{matrix} \\ \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \end{array} \begin{array}{l} \leftarrow \text{Row } 1 \\ \leftarrow \text{Row } 2 \\ \\ \leftarrow \text{Row } i \\ \\ \leftarrow \text{Row } m \end{array} \Bigg\downarrow m \text{ rows}$$

Note that m and n are positive integers.

The following are some examples of matrices :

$$A=\begin{bmatrix} 2 & 0 & -1 \\ 1 & 4 & 5 \\ 9 & -8 & 6 \end{bmatrix},\ B=\begin{bmatrix} 7 & -9 & 1.2 & 0 \\ \sin\frac{x}{4} & 2 & x^2 & 4 \\ \cos\frac{x}{2} & 1 & 3 & -6 \end{bmatrix}, \text{and } C=\begin{bmatrix} 1 & 5 & -7 \\ 3.4 & \frac{1}{2} & \sqrt{2} \\ e^2 & -3 & 4 \\ \sqrt{5} & 2 & a \end{bmatrix}.$$

In a matrix, the horizontal lines of elements are known as **rows** and the vertical lines of elements are known as **columns**. Thus A has 3 rows and 3 columns, B has 3 rows and 4 columns, and C has 4 rows and 3 columns.

Definition 7.1

If a matrix A has m rows and n columns then the **order or size** of the matrix A is defined to be $m \times n$ (read as m by n).

The objects $a_{11}, a_{12}, \ldots, a_{mn}$ are called **elements or entries** of the matrix $A=[a_{ij}]_{m\times n}$. The element a_{ij} is common to i^{th} row and j^{th} column and is called $(i,j)^{\text{th}}$ element of A. Observe that the ith row and j^{th} column of A are $1\times n$ and $m\times 1$ matrices respectively and are given by $[a_{i1}\ \ a_{i2}\ \ \ldots\ \ a_{in}]$ and $\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$

We shall now visualize the representation and construction of matrices for simplifying day-to-day problems.

Illustration 7.1

Consider the marks scored by a student in different subjects and in different terminal examinations. They are exhibited in a tabular form as given below :

	Tamil	English	Mathematics	Science	Social Science
Exam 1	48	71	80	62	55
Exam 2	70	68	91	73	60
Exam 3	77	84	95	82	62

This tabulation represents the above information in the form of matrix. What does the entry in the third row and second column represent?

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